



香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

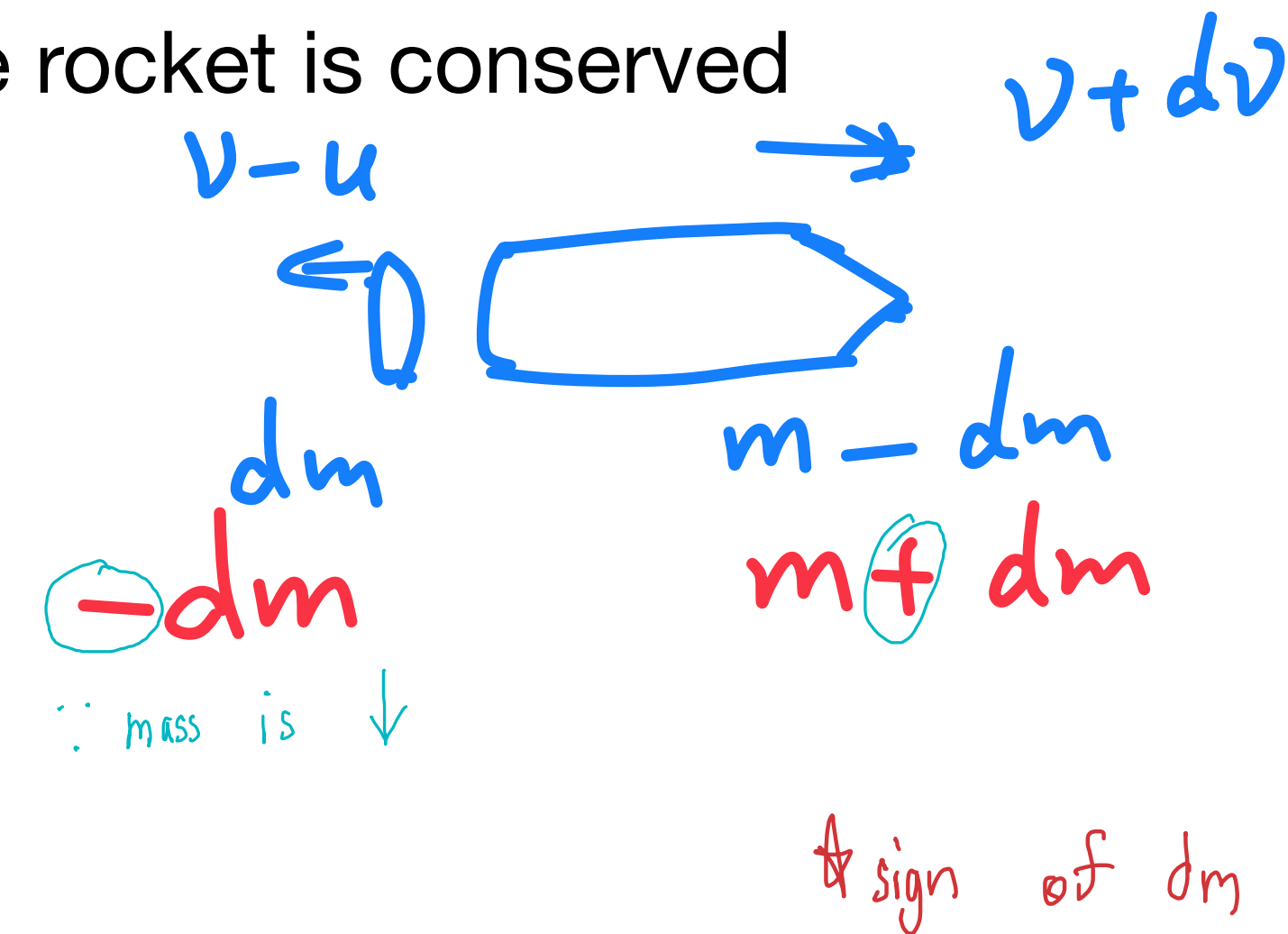
PHY1001 Mechanics

Jingqiang Ye
School of Science and Engineering (SSE)
February 25th, 2025

Rocket Propulsion: Varying Mass

- Assume the rocket is in space and has no gravity.
- Fuel is rejected from the rocket with a relative velocity u (v_{rel} in some textbook) to the rocket.
- The momentum of the system of the rejected fuel and the rocket is conserved
- First rocket equation: $Ru = ma$, where

- ▶ $R = \frac{dm}{dt}$ is the fuel consumption rate in kg/s
 $\hookrightarrow$ fre
- ▶ m is the instantaneous mass of the rocket
- ▶ a is the instantaneous acceleration



$$(m + dm)(v + dv) + (-dm)(v - u) = mv$$

$$m v + m dv + v dm - v dm + u dm = m v$$

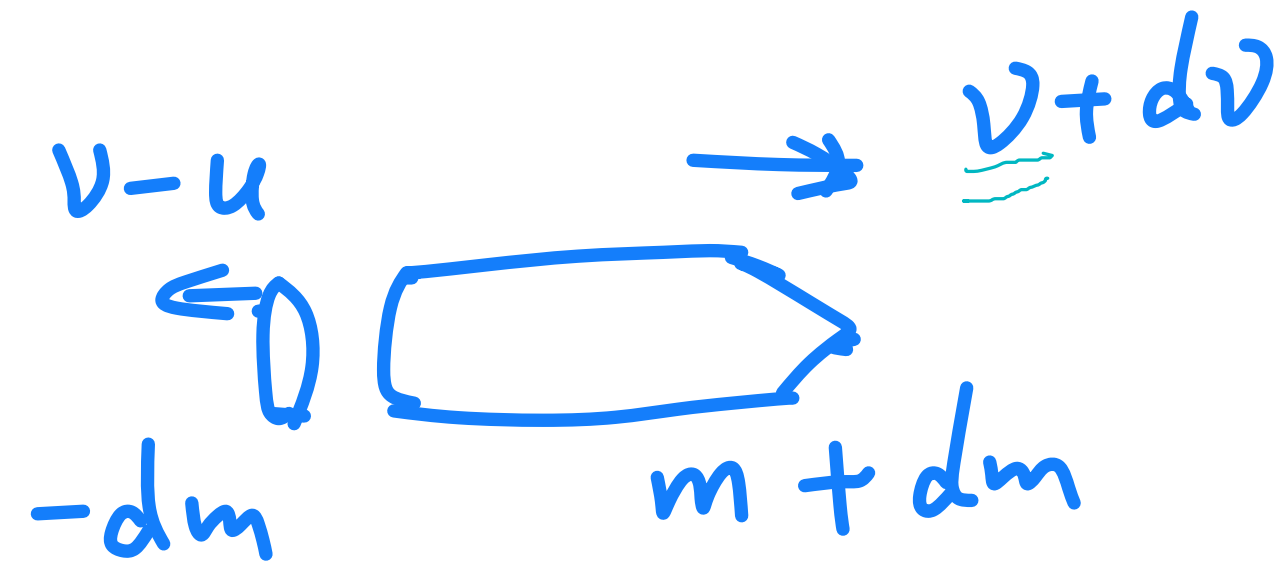
$$m dv + u dm = 0$$

$$m \frac{dv}{dt} + u \frac{dm}{dt} = 0$$

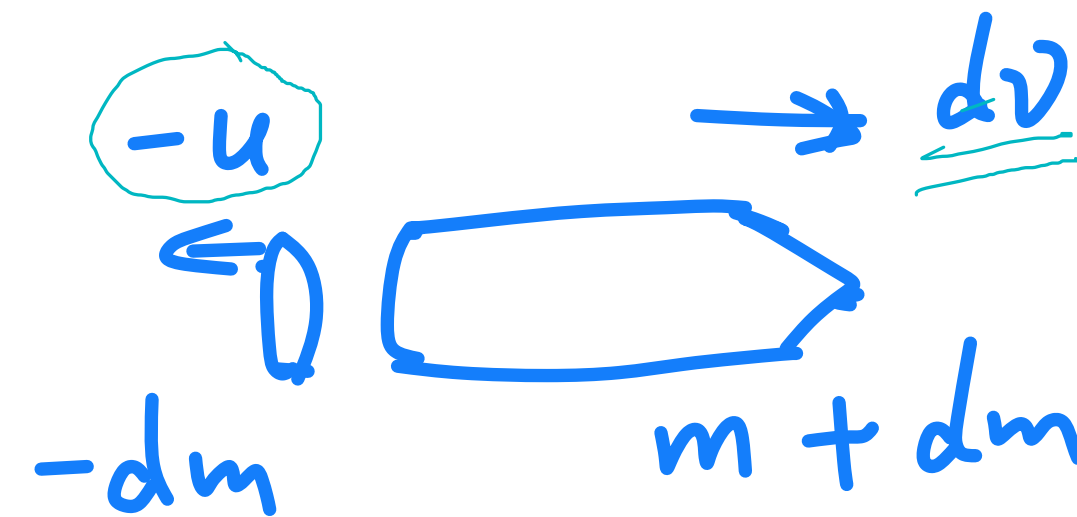
$$ma - uR = 0$$

In the Frame of the Rocket

Frame of the space



Frame of the rocket



$$(m+dm)dv + (-dm)(-u) = 0$$
$$mdv + udm = 0$$

Center of Mass

- The center of mass is the mass-weighted center of a system

- mathematically: $x_{\text{com}} = \frac{\sum_i m_i x_i}{\sum_i m_i}$ (1D case)

- For higher dimensions: $\vec{r}_{\text{com}} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$

- The total momentum of a system $\vec{P} = M \vec{v}_{\text{com}}$, where
 - $M = \sum_i m_i$
 - $\hookrightarrow$ if P conserve
 - $\hookrightarrow$ fixed $M \rightarrow$ constant

$$\vec{v}_{\text{com}} = \frac{d\vec{x}_{\text{com}}}{dt}$$

$$= \frac{\sum_i m_i \frac{d\vec{x}_i}{dt}}{\sum_i m_i}$$

- $\vec{v}_{\text{com}}$ is the velocity of center of mass
- Useful for momentum conservation calculation

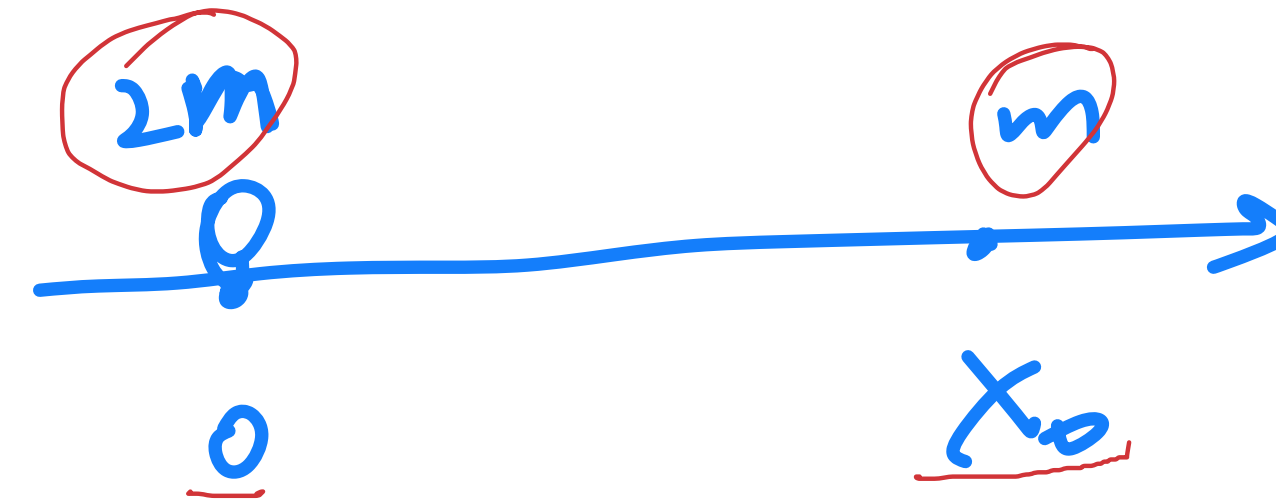
Exercise

A ball of mass $2m$ and a ball m are on a horizontal smooth surface. The $2m$ ball is at the origin, while the m ball is at $x = x_0$, as shown in the plot.

a) Find the center of mass of this two-ball system.

$$x_{\text{com}} = \frac{2m \cdot 0 + m x_0}{3m}$$

$$= \frac{x_0}{3}$$

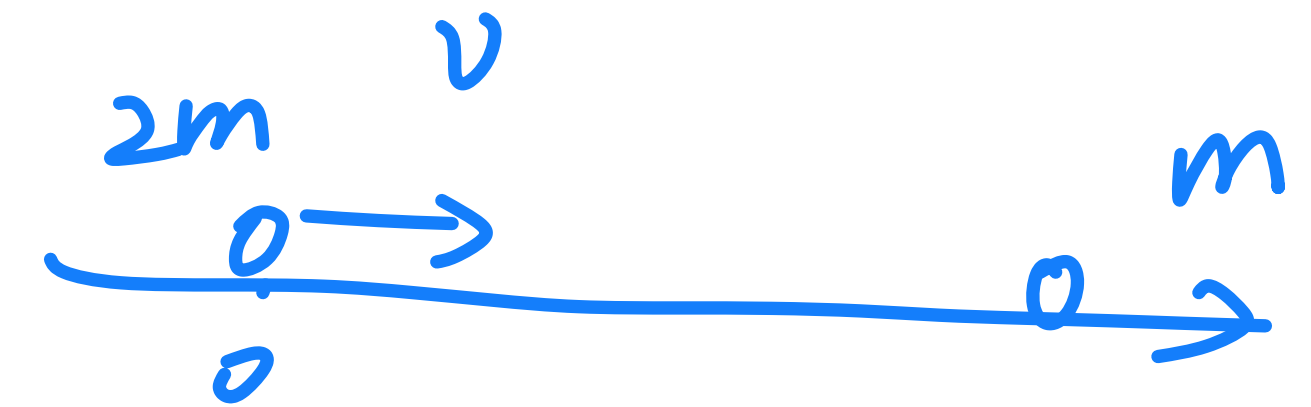


Exercise

A ball of mass $2m$ collides elastically with a ball m on a horizontal smooth surface. The $2m$ ball has a velocity of v , while the m ball is at rest before the collision.

b) Find the velocity of the two balls after the collision using the lab frame.

$$p: \begin{cases} 2mv = 2m \cdot v_1 + m v_2 \\ k: \begin{cases} \frac{1}{2} 2mv^2 = \frac{1}{2} 2m v_1^2 + \frac{1}{2} m v_2^2 \end{cases} \end{cases}$$



$$\begin{cases} 2(v - v_1) = v_2 \\ 2(v - v_1)(v + v_1) = v_2^2 \end{cases}$$

$$v + v_1 = v_2$$

$$2(v - v_1) = v + v_2$$

$$v_1 = \frac{v}{3}$$

$$v_2 = \frac{4}{3} v$$

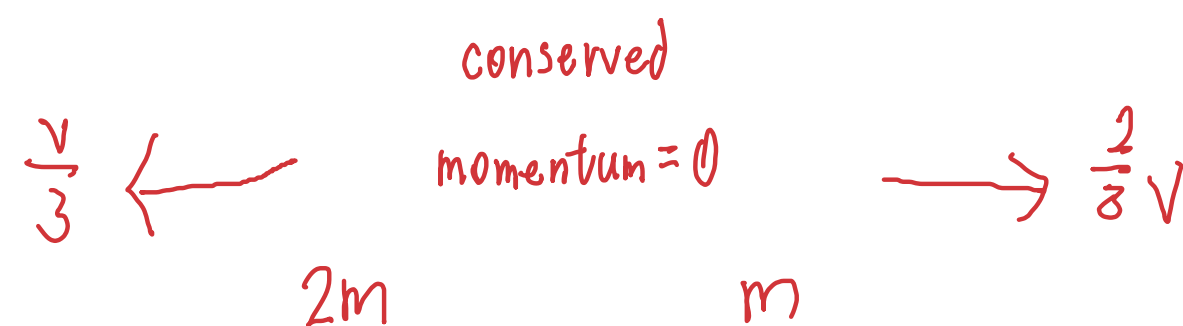
Exercise

A ball of mass $2m$ collides elastically with a ball m on a horizontal smooth surface. The $2m$ ball has a velocity of v , while the m ball is at rest before the collision.

c) Find the velocity of the two balls after the collision using the frame of center of mass.

$$V_{\text{com}} = \frac{2m \cdot v + 0}{3m} = \frac{2}{3}v$$

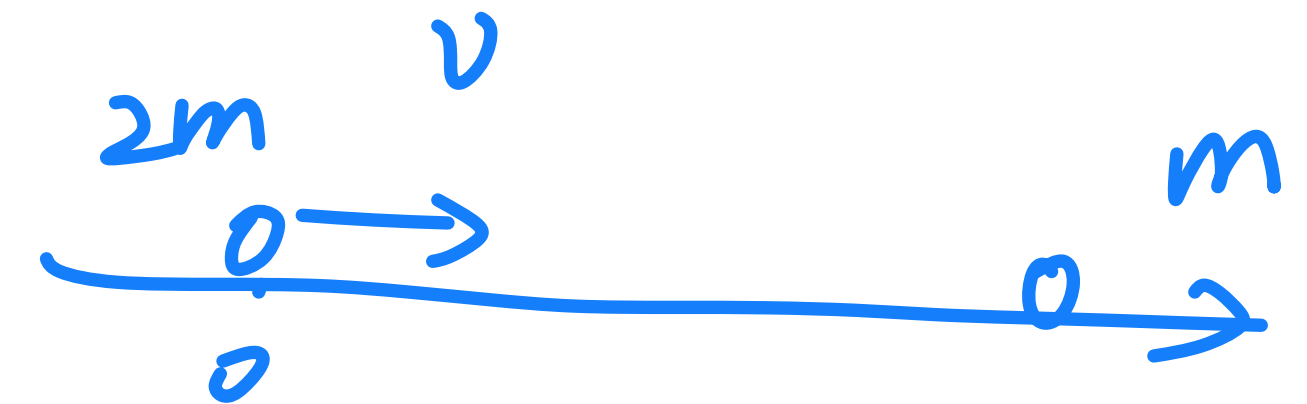
COM frame



lab frame

$$V_1 = \frac{2}{3}v + \left(-\frac{1}{3}v\right) = \frac{v}{3}$$

$$V_2 = \frac{2}{3}v + \frac{2}{3}v = \frac{4}{3}v$$

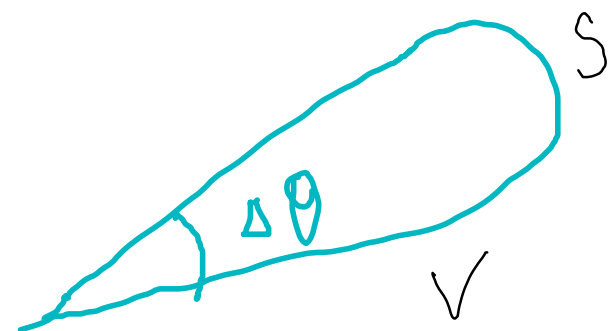


Angular Displacement

- Angular displacement $\Delta\theta = \frac{s}{r}$, where s is the length of the arc, r is the radius of the circle
 - Angular displacement is not a vector, unless it is infinitesimal
- Unit: rad (radian); it is dimensionless.

$$\Delta\theta = \frac{s}{r}$$

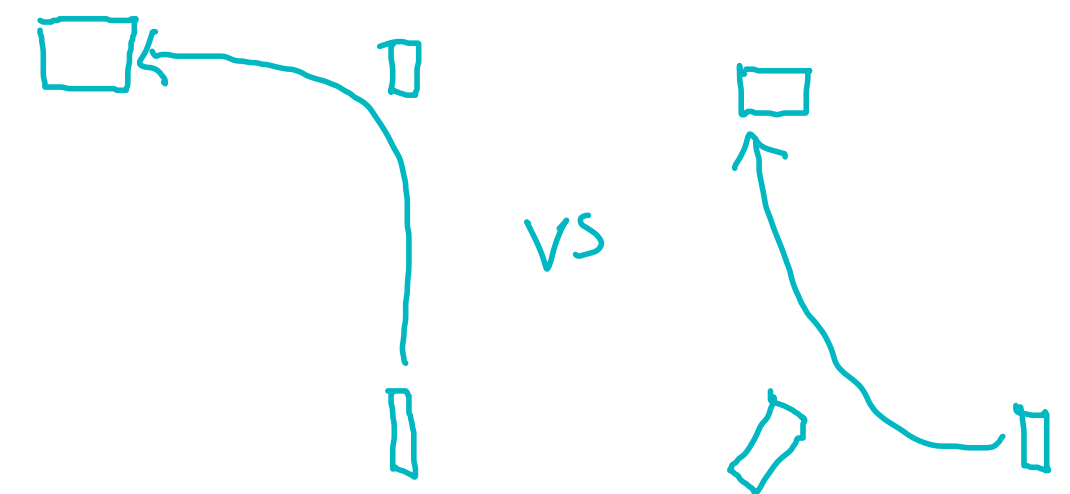
$$\left[\frac{s}{r}\right] = \frac{\text{m}}{\text{m}} = 1$$



full circle $\Delta\theta = 2\pi = \frac{2\pi r}{r} = 22$

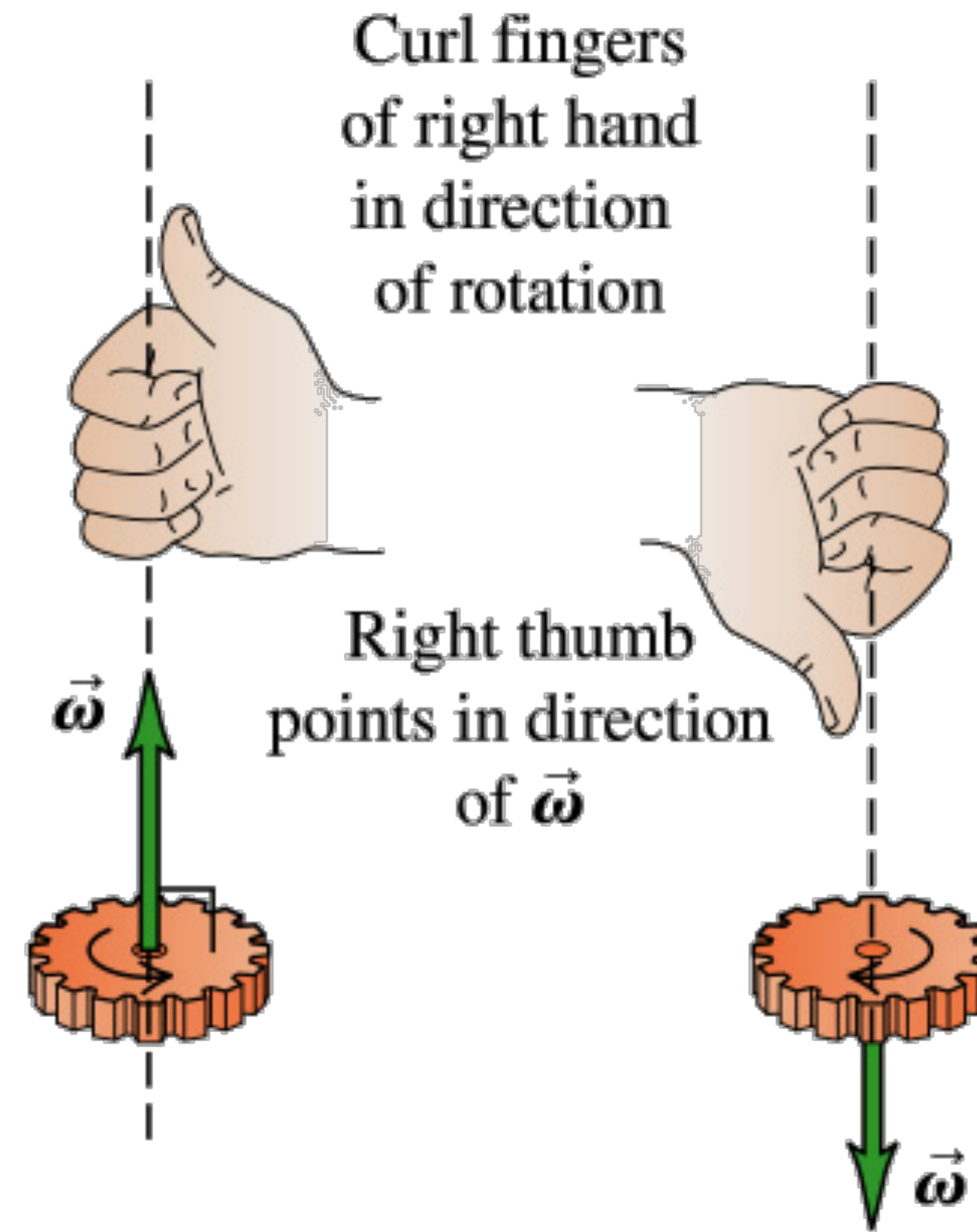
$$\vec{A} + \vec{B} = \vec{B} + \vec{A}$$

$d\vec{\theta}$ is a vector



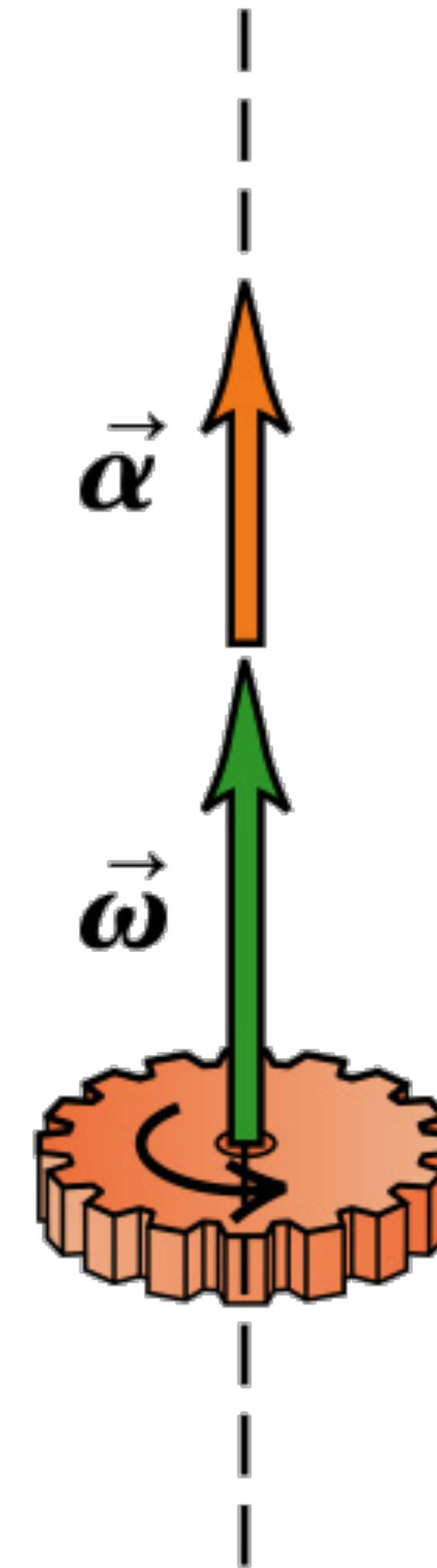
Angular Velocity

- $\omega = \frac{d\theta}{dt}$
 - Unit: rad/s
 - Direction: Right hand rule
- Vector form: $\vec{\omega}_z = \frac{d\theta}{dt} \hat{k}$
 - ↳ vector!

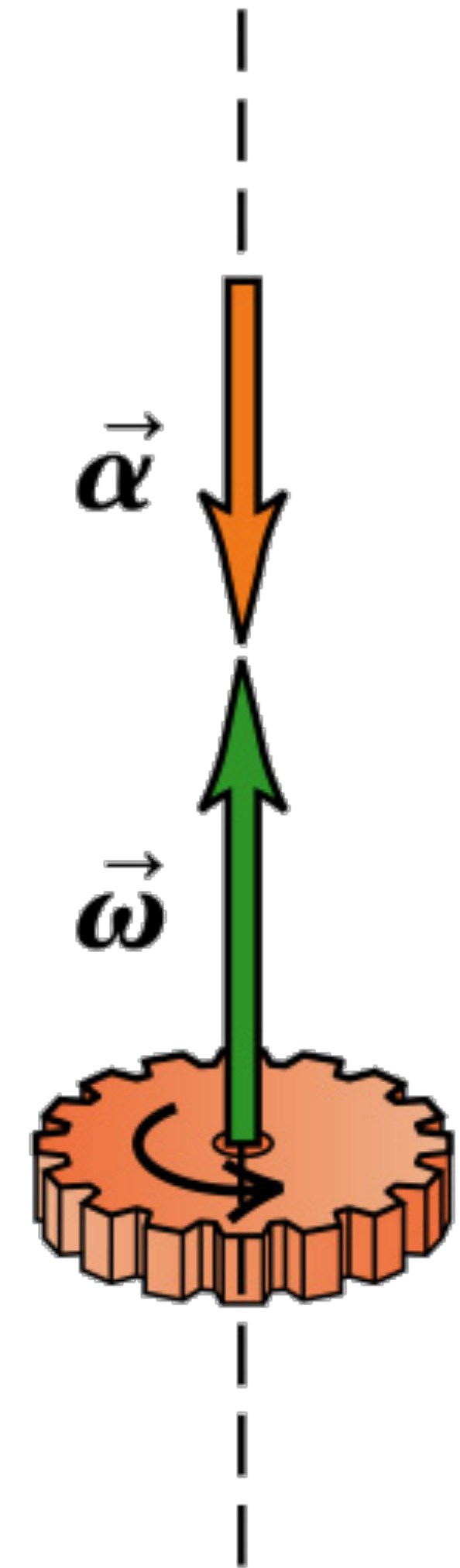


Angular Acceleration

- Angular acceleration $\vec{\alpha} = \frac{d\vec{\omega}}{dt} = \frac{d^2\theta}{dt^2}\hat{k}$
- Unit: rad/s^2
- If $\vec{\alpha}$ and $\vec{\omega}$ are in the same direction $\rightarrow$ rotation accelerates.
- $\vec{\alpha}$ and $\vec{\omega}$ opposite $\rightarrow$ rotation decelerates



Speeding up



Slowing up

Linear & Rotation Motion Compared

Rotation

$$\theta$$

$$\vec{\omega} = \frac{d\theta}{dt} \hat{k}$$

$$\vec{\alpha} = \frac{d\vec{\omega}}{dt} = \frac{d^2\theta}{dt^2} \hat{k}$$

Linear

$$x$$

$$\vec{v} = \frac{d\vec{x}}{dt}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{x}}{dt^2}$$

Linear & Rotation Motion Compared w/ Constant Acceleration

Straight-Line Motion with Constant Linear Acceleration

$$a_x = \text{constant}$$

$$v_x = v_{0x} + a_x t \quad (2.8)$$

$$x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2 \quad (2.12)$$

$$v_x^2 = v_{0x}^2 + 2a_x(x - x_0) \quad (2.13)$$

$$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t \quad (2.14)$$

Fixed-Axis Rotation with Constant Angular Acceleration

$$\alpha_z = \text{constant}$$

$$\omega_z = \omega_{0z} + \alpha_z t \quad (9.7)$$

$$\theta = \theta_0 + \omega_{0z}t + \frac{1}{2}\alpha_z t^2 \quad (9.11)$$

$$\omega_z^2 = \omega_{0z}^2 + 2\alpha_z(\theta - \theta_0) \quad (9.12)$$

$$\theta - \theta_0 = \frac{1}{2}(\omega_{0z} + \omega_z)t \quad (9.10)$$

Exercise

A wheel rotates with a constant angular acceleration $\alpha = 3.50 \text{ rad/s}^2$, and its angular velocity $\omega_0 = 2 \text{ rad/s}$ at $t=0$.

- a) What is the angular displacement it goes through between $t=0$ and $t=2 \text{ s}$?
b) What is the angular speed at $t=2 \text{ s}$?

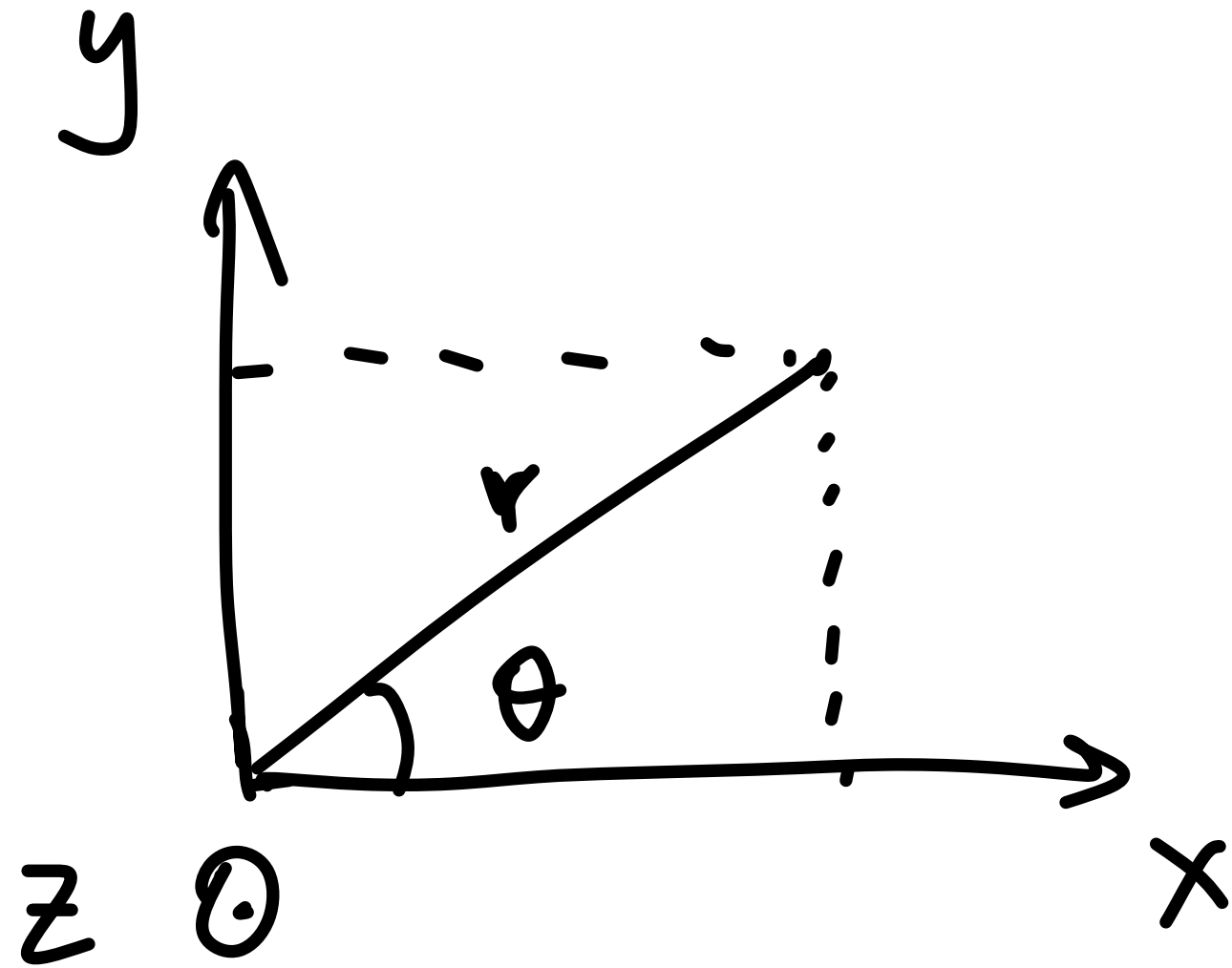
$$a) \Delta \theta = 0 + 2 \cdot 2 + \frac{1}{2} \cdot 3.5 \cdot 2^2$$

$$= 11.0 \text{ rad} = 11 \cdot \frac{1}{2\pi} \text{ rev} = 1.8 \text{ rev}$$

$$\star \boxed{1 \text{ rev} = 2\pi \text{ rad}}$$

$$b) \omega = 2 + 3.5 \cdot 2 \\ = 9.0 \text{ rad/s}$$

Polar Coordinates



- $\hat{r} \times \hat{\theta} = \hat{k}$
- $\hat{\theta} \times \hat{k} = \hat{r}$
- $\hat{k} \times \hat{r} = \hat{\theta}$

Vector Form

$$V = r\omega$$

$$a_{\text{tan}} = \frac{dv}{dt} = r \frac{d\omega}{dt} = r\alpha$$

$$a_{\text{rad}} = \frac{v^2}{r} = \frac{\omega^2 r^2}{r} = \omega^2 r$$

$$\vec{v} = \vec{\omega} \times \vec{r}$$

$$\vec{a}_{\text{tan}} = \vec{\alpha} \times \vec{r}$$

$$\vec{a}_{\text{rad}} = -\omega^2 \vec{r}$$

Alternately

$$\vec{a}_{\text{rad}} = \vec{\omega} \times \vec{v} \quad (\text{HW})$$

Rigid Body

- Rigid body is an idealized model. It represents a body that is non-deformable.
 - If a rigid body rotates around certain axis, the angular velocity ω is the same everywhere in the rigid body!
- We will use rigid body to study rotation unless specified.

Rotational Kinetic Energy

- $K = \frac{1}{2}I\omega^2$, where

$$K = \frac{1}{2}mv^2$$

- K is the rotational kinetic energy
- I is the **moment of inertia**
- ω is the angular velocity

$$K = \sum_i \frac{1}{2} m_i v_i^2$$

$$= \sum_i \frac{1}{2} m_i (\omega r_i)^2$$

$$= \frac{1}{2} \omega^2 I$$

- $I = \sum_i m_i r_i^2$ for discrete mass distribution, or $I = \int r^2 dm$ for continuous mass distribution
- Unit: $\text{kg} \cdot \text{m}^2$

Mass vs. Moment of Inertia

Quantity	Mass	Mol
Definition	$m = \sum_i m_i$	$I = \sum_i m_i r_i^2$
Scalar or vector?	Scalar	Scalar
Unit	kg	$\text{kg} \cdot \text{m}^2$
Kinetic energy	$K = \frac{1}{2}mv^2$	$K = \frac{1}{2}I\omega^2$

Center of Mass

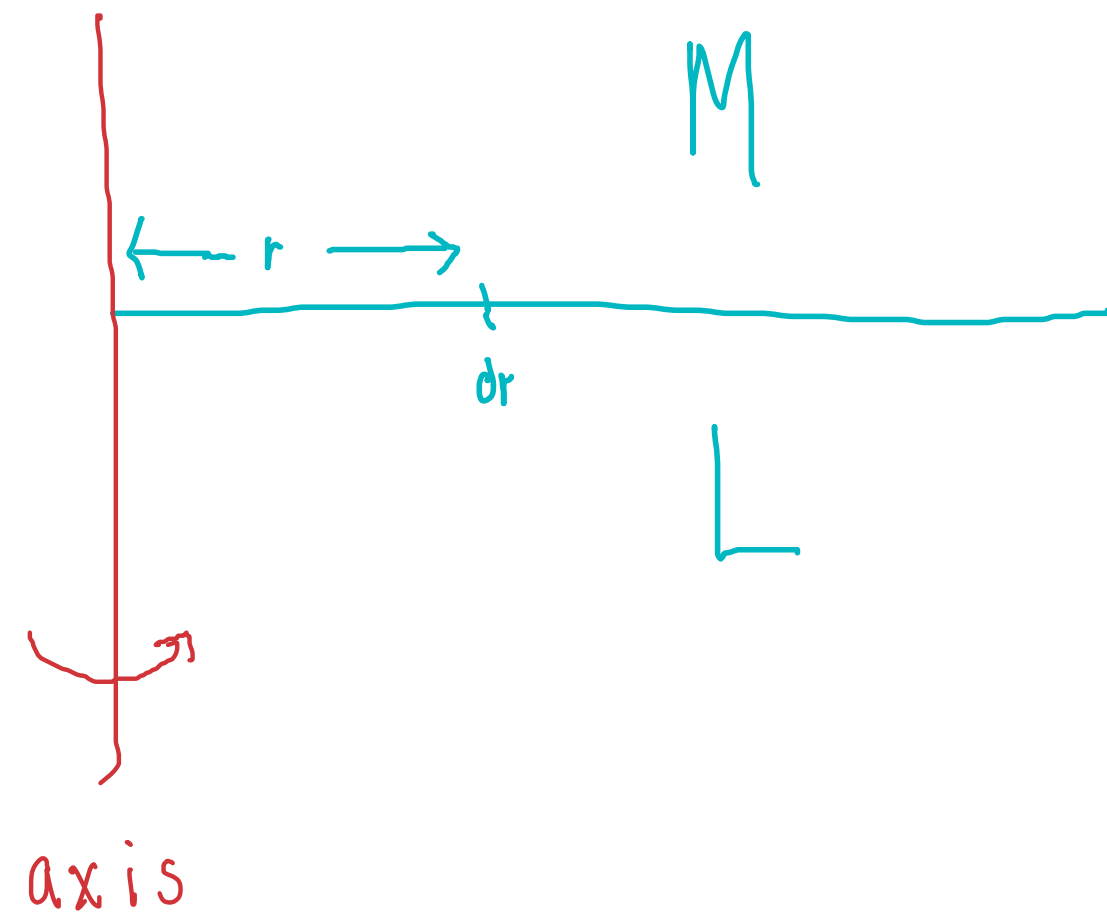
$$\vec{r}_{\text{CM}} = \frac{\sum_i m_i \vec{r}_i}{\sum_i m_i}$$

$$I = \int r^2 dm$$

r distance
 $axis$
 dm

Exercise

Find the moment of inertia for a ^{1D ← thin rod} ~~solid ball~~ of mass M and ^{length L} ~~radius R~~ , if it rotates around an axis through its ^{end} ~~center~~. Suppose its density is uniform.



linear density $\rho = \frac{M}{L} \text{ kg/m}$

$$I = \int r^2 dm$$

$$dm = dr \cdot \frac{M}{L}$$

$$\begin{aligned} I &= \int_0^L r^2 \cdot dr \cdot \frac{M}{L} \\ &= \frac{M}{L} \cdot \frac{r^3}{3} \Big|_0^L = \frac{M}{L} \cdot \frac{L^3}{3} \\ &= \frac{ML^2}{3} \end{aligned}$$